

RAJARATA UNIVERSITY OF SRI LANKA

FACULTY OF TECHNOLOGY

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

ICT 3219 – Mobile Application Development

Practical Sheet – 02

Understand the purpose and use cases of different Android layouts

Task 01 – Relative Layout

Relative Layout positions each child relative to the parent or to other sibling views — using rules like "below this," "to the right of that," "aligned with this view's top," etc.

Create a New Layout XML File

1. Right-click on res > layout.
2. Click **New > Layout Resource File**.
3. Name the file, e.g., relative_layout.xml.
4. Choose Root Element: Relative Layout, etc. based on layout type.
5. Click **OK**.

relative_layout.xml

1. Go to res > layout > relative_layout.xml.
2. Switch to **Code view** (not Design view).
3. **Replace the existing code** with the provided Relative Layout code.

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="24dp"
    android:background="#E3F2FD">

    <!-- Dashboard Title -->
    <TextView
        android:id="@+id/title"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="My Dashboard"
        android:textSize="28sp"
        android:textStyle="bold"
        android:textColor="#0D47A1" />

    <!-- Settings Button aligned to end of title -->
    <Button
        android:id="@+id/settings"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Settings"
```

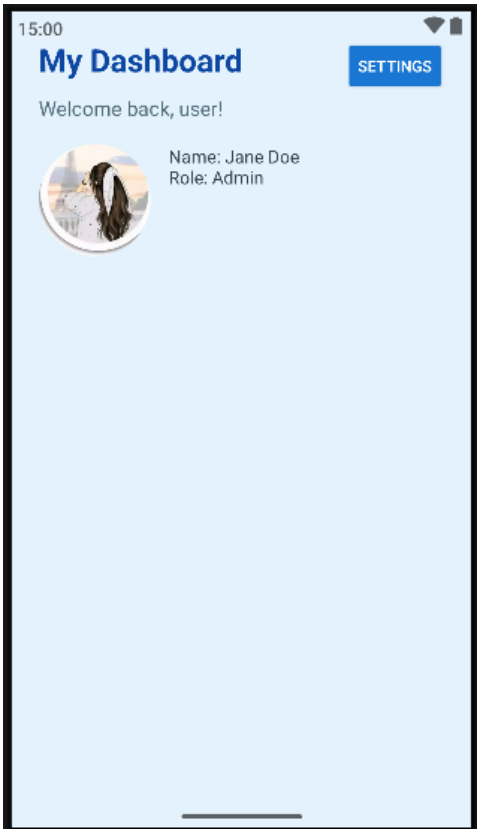
```
        android:textColor="#FFFFFF"
        android:backgroundTint="#1976D2"
        android:layout_alignBaseline="@id/title"
        android:layout_alignParentEnd="true" />

    <!-- Subtitle under the title -->
    <TextView
        android:id="@+id/subtitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome back, user!"
        android:textSize="18sp"
        android:textColor="#546E7A"
        android:layout_below="@id/title"
        android:layout_marginTop="12dp" />

    <!-- Profile Image below subtitle -->
    <ImageView
        android:id="@+id/profile_image"
        android:layout_width="100dp"
        android:layout_height="100dp"
        android:layout_below="@id/subtitle"
        android:layout_marginTop="20dp"
        android:src="@drawable/profilepicture"
        android:layout_alignParentStart="true"
        android:background="@drawable/circle_bg"
        android:scaleType="centerCrop"
        android:clipToOutline="true" />

    <!-- User Info to the right of image -->
    <TextView
        android:id="@+id/user_info"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name: Jane Doe\nRole: Admin"
        android:textSize="16sp"
        android:textColor="#37474F"
        android:layout_toEndOf="@id/profile_image"
        android:layout_alignTop="@id/profile_image"
        android:layout_marginStart="16dp" />

</RelativeLayout>
```



Task 02 – Table Layout

Table Layout arranges children into **rows and columns**, like an HTML table or spreadsheet. It's made up of Table Row elements — each Table Row represents one row, and each child inside that row becomes one column/cell.

Create a New Layout XML File

6. Right-click on res > layout.
7. Click **New > Layout Resource File**.
8. Name the file, e.g., table_layout.xml.
9. Choose Root Element: Table Layout, etc. based on layout type.
10. Click **OK**.

table_layout.xml

4. Go to res > layout > table_layout.xml.
5. Switch to **Code view** (not Design view).
6. **Replace the existing code** with the provided Table Layout code.

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#FAFAFA"
    android:padding="16dp"
    android:orientation="vertical">

    <!-- Title -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="📅 My Weekly Timetable"
        android:textSize="24sp"
        android:textStyle="bold"
        android:textColor="#3E7D80"
        android:layout_gravity="center_horizontal"
        android:layout_marginBottom="16dp" />
```

```
<!-- Timetable Table -->
<TableLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:stretchColumns="1"
    android:background="#00000000"
    android:elevation="4dp"
    android:padding="12dp">

    <!-- Header Row -->
    <TableRow android:background="#3E7D80">
        <TextView
            android:text="Time"
            android:textStyle="bold"
            android:textSize="16sp"
            android:textColor="#FFFFFF"
            android:padding="10dp" />

        <TextView
            android:text="Subject"
            android:textStyle="bold"
            android:textSize="16sp"
            android:textColor="#FFFFFF"
            android:padding="10dp" />
    </TableRow>

    <!-- Row 1 -->
    <TableRow android:background="#F1F9E9">
        <TextView
            android:text="08:00 - 09:00"
            android:textSize="15sp"
            android:padding="10dp"
            android:textColor="#000000" />

        <TextView
            android:text="Mathematics"
            android:textSize="15sp"
            android:padding="10dp"
            android:textColor="#000000" />
    </TableRow>

    <!-- Row 2 -->
    <TableRow android:background="#E3E2F0">
        <TextView
            android:text="09:00 - 10:00"
            android:textSize="15sp"
            android:padding="10dp"
            android:textColor="#000000" />

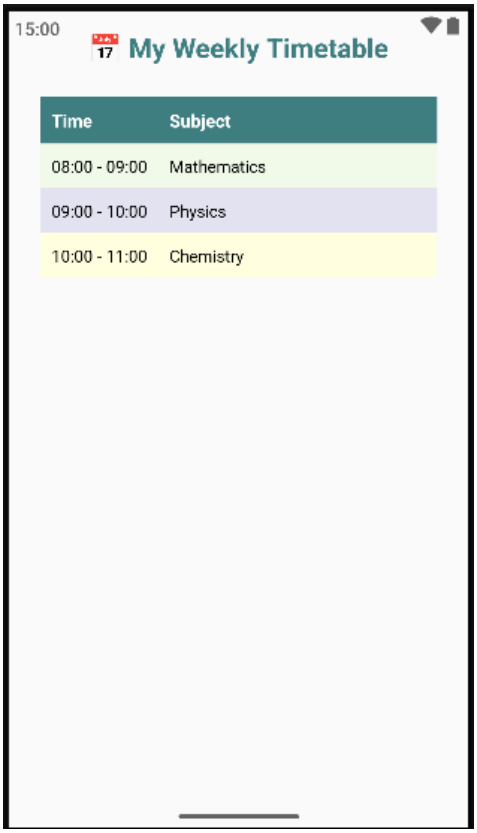
        <TextView
            android:text="Physics"
            android:textSize="15sp"
            android:padding="10dp"
            android:textColor="#000000" />
    </TableRow>

    <!-- Row 3 -->
    <TableRow android:background="#FFFFE0">
        <TextView
            android:text="10:00 - 11:00"
            android:textSize="15sp"
            android:padding="10dp"
            android:textColor="#000000" />
```

```
            <TextView
                android:text="Chemistry"
                android:textSize="15sp"
                android:padding="10dp"
                android:textColor="#000000" />
        </TableRow>

    </TableLayout>

</LinearLayout>
```

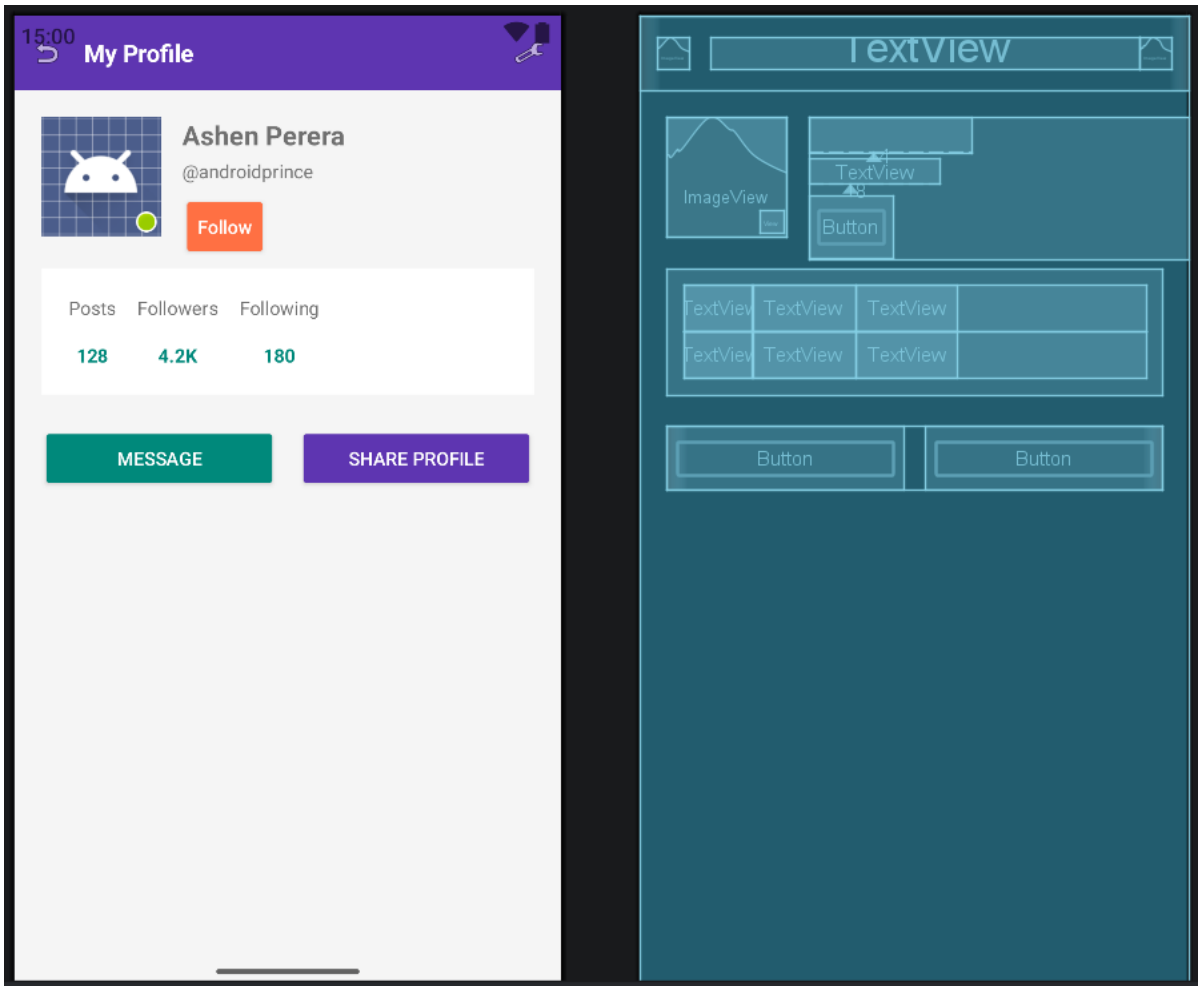


Activity 01

Design and implement an Android screen that combines at least four of the layouts you've learned — LinearLayout, RelativeLayout, FrameLayout, TableLayout, and ConstraintLayout into a single, cohesive UI.

Design a Profile Dashboard screen as the image that includes the following sections

1. Header bar — app title with icons on either side
2. Profile picture with a status badge overlapping its corner
3. Name and username block positioned beside the profile picture
4. A stats section (e.g. Posts / Followers / Following) organized into a clean row-and-column grid
5. Action buttons at the bottom (e.g. Message, Follow, Share)




```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/background_grey">

    <!-- ============ 1. HEADER : LinearLayout ============ -->
    <LinearLayout
        android:id="@+id/headerBar"
        android:layout_width="0dp"
        android:layout_height="56dp"
```

```
        android:orientation="horizontal"
        android:gravity="center_vertical"
        android:background="@color/header_purple"
        android:paddingStart="12dp"
        android:paddingEnd="12dp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent">

        <ImageView
            android:layout_width="24dp"
            android:layout_height="24dp"
            android:src="@android:drawable/ic_menu_revert"
            android:contentDescription="Back" />

        <TextView
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:layout_marginStart="16dp"
            android:text="My Profile"
            android:textColor="#FFFFFF"
            android:textSize="18sp"
            android:textStyle="bold" />

        <ImageView
            android:layout_width="24dp"
            android:layout_height="24dp"
            android:src="@android:drawable/ic_menu_manage"
            android:contentDescription="Settings" />
    </LinearLayout>

    <!-- ============ 2. AVATAR + BADGE : FrameLayout ============ -->
    <FrameLayout
        android:id="@+id/avatarFrame"
        android:layout_width="90dp"
        android:layout_height="90dp"
        android:layout_marginTop="20dp"
        android:layout_marginStart="20dp"
        app:layout_constraintTop_toBottomOf="@id/headerBar"
        app:layout_constraintStart_toStartOf="parent">

        <ImageView
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:src="@android:drawable/sym_def_app_icon"
            android:background="#DDDDDD"
            android:contentDescription="Avatar" />

        <!-- Online status badge, overlapping bottom-right corner of the avatar -->
        <View
            android:layout_width="18dp"
            android:layout_height="18dp"
            android:layout_gravity="bottom|end"
            android:background="@android:drawable/presence_online"
            android:layout_marginEnd="2dp"
            android:layout_marginBottom="2dp" />
    </FrameLayout>

    <!-- ============ 3. NAME CARD : RelativeLayout ============ -->
    <RelativeLayout
        android:id="@+id/infoCard"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginStart="16dp"
```

```
        android:layout_marginTop="20dp"
        app:layout_constraintTop_toBottomOf="@id/headerBar"
        app:layout_constraintStart_toEndOf="@id/avatarFrame"
        app:layout_constraintEnd_toEndOf="parent">

        <TextView
            android:id="@+id/nameText"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Ashen Perera"
            android:textSize="20sp"
            android:textStyle="bold" />

        <TextView
            android:id="@+id/usernameText"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_below="@id/nameText"
            android:layout_marginTop="4dp"
            android:text="\@androidprince"
            android:textColor="#777777" />

        <Button
            android:id="@+id/followBtn"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_below="@id/usernameText"
            android:layout_marginTop="8dp"
            android:text="Follow"
            android:textAllCaps="false"
            android:backgroundTint="@color/accent_orange"
            android:textColor="@color/card_white"
            style="?android:attr/buttonStyleSmall" />
    </RelativeLayout>

    <!-- ============ 4. STATS : TableLayout ============ -->
    <TableLayout
        android:id="@+id/statsTable"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"
        android:layout_marginStart="20dp"
        android:layout_marginEnd="20dp"
        android:background="@color/card_white"
        android:padding="12dp"
        app:layout_constraintTop_toBottomOf="@id/avatarFrame"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent">

        <TableRow>
            <TextView
                android:layout_column="0"
                android:text="Posts"
                android:padding="8dp"
                android:gravity="center"
                android:textColor="@color/text_grey" />
            <TextView
                android:text="Followers"
                android:padding="8dp"
                android:gravity="center"
                android:textColor="@color/text_grey" />
            <TextView
                android:text="Following"
                android:padding="8dp"
                android:gravity="center"
```

```
                android:textColor="@color/text_grey" />
        </TableRow>

        <TableRow>
            <TextView
                android:text="128"
                android:padding="8dp"
                android:gravity="center"
                android:textStyle="bold"
                android:textColor="@color/teal_primary" />
            <TextView
                android:text="4.2K"
                android:padding="8dp"
                android:gravity="center"
                android:textStyle="bold"
                android:textColor="@color/teal_primary" />
            <TextView
                android:text="180"
                android:padding="8dp"
                android:gravity="center"
                android:textStyle="bold"
                android:textColor="@color/teal_primary" />
        </TableRow>
    </TableLayout>

    <!-- ============ 5. ACTION ROW : LinearLayout ============ -->
    <LinearLayout
        android:id="@+id/actionRow"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_marginTop="24dp"
        android:layout_marginStart="20dp"
        android:layout_marginEnd="20dp"
        android:weightSum="2"
        app:layout_constraintTop_toBottomOf="@id/statsTable"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent">

        <Button
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:layout_marginEnd="8dp"
            android:text="Message"
            android:backgroundTint="@color/teal_primary"
            android:textColor="@color/card_white" />

        <Button
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:layout_marginStart="8dp"
            android:text="Share Profile"
            android:backgroundTint="@color/header_purple"
            android:textColor="@color/card_white" />
    </LinearLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```